WORKED SOLUTION

STEP II 2025, Question 1

Pure • Piecewise Functions

STEP II 2025, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

A function built from Min is just an ordinary function given by different formulas on different intervals. So the first job every time is to find where the two arguments cross, write the function piecewise, and then work interval by interval, keeping only the answers that land in the interval they came from. At the joins the function stays continuous but usually has a corner, and that corner is what makes the point in (iv) a maximum without the derivative being zero.

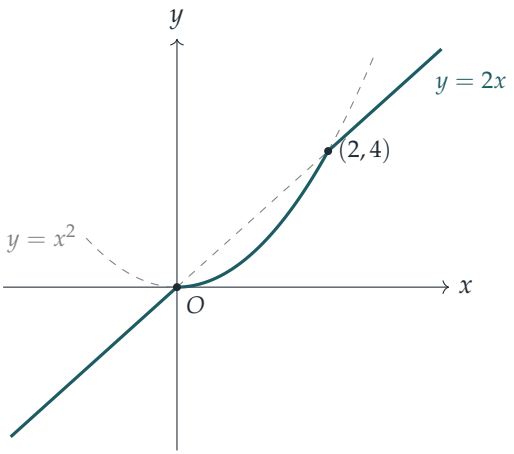
Solution

Part (i)

We have $x^2 \leq 2x \iff x(x-2) \leq 0 \iff 0 \leq x \leq 2$. So

$$\mathrm{Min}(x^2, 2x) = \begin{cases} x^2 & 0 \leq x \leq 2, \\ 2x & x < 0 \text{ or } x > 2. \end{cases}$$

The graph follows the line $y = 2x$ except between $x = 0$ and $x = 2$, where it dips onto the parabola, which lies below its chord there. Both straight pieces lie on the *same* line. There are corners at $(0, 0)$, where the parabola has gradient 0 but the line has gradient 2, and at $(2, 4)$, where the gradients are 4 and 2.



Part (ii)

Use the two pieces from (i).

- For $0 \leq x \leq 2$: $2x^2 = 5x - 3 \iff 2x^2 - 5x + 3 = 0 \iff (2x-3)(x-1) = 0$, so $x = 1$ or $x = \frac{3}{2}$. Both lie in $[0, 2]$, so both count.
- For $x < 0$ or $x > 2$: $4x = 5x - 3 \iff x = 3$, which satisfies $x > 2$, so it counts.

The solutions are $x = 1$, $x = \frac{3}{2}$ and $x = 3$.

Part (iii)

We also need the second function: $x^3 \leq 4x \iff x(x-2)(x+2) \leq 0 \iff x \leq -2$ or $0 \leq x \leq 2$. Putting the two together,

$$\text{Min}(x^2, 2x) + \text{Min}(x^3, 4x) = \begin{cases} x^3 + 2x & x \leq -2, \\ 6x & -2 < x < 0, \\ x^3 + x^2 & 0 \leq x \leq 2, \\ 6x & x > 2. \end{cases}$$

$m = 2$.

- $x \leq -2$: $x^3 + 2x = 2x$ gives $x = 0$, not in the interval.
- $-2 < x < 0$ and $x > 2$: $6x = 2x$ gives $x = 0$, in neither interval.
- $0 \leq x \leq 2$: $x^3 + x^2 - 2x = x(x+2)(x-1) = 0$ gives $x = 0$ or $x = 1$ (we reject $x = -2$).

So for $m = 2$ the solutions are $x = 0$ and $x = 1$.

? **Ask yourself** What happens to the equation $6x = 6x$?

It is true for *every* x, so when $m = 6$ whole intervals are solutions, not just isolated points.

$m = 6$**.**

- $x \le -2$: $x^3 + 2x = 6x \iff x(x-2)(x+2) = 0$, and only $x = -2$ is in the interval.
- $-2 < x < 0$ and $x > 2$: $6x = 6x$ holds identically, so every x in these intervals is a solution.
- $0 \le x \le 2$: $x^3 + x^2 - 6x = x(x+3)(x-2) = 0$ gives $x = 0$ or $x = 2$ (we reject $x = -3$).

Combining, for $m = 6$ the solution set is

$$-2 \le x \le 0 \quad \text{or} \quad x \ge 2.$$

Part (iv)

Here $x^3 \le x^2 \iff x^2(x-1) \le 0 \iff x \le 1$ (the factor x^2 is never negative, and $x = 0$ gives equality). So

$$y = \begin{cases} 2x^3 - 5x & x \le 1, \\ 2x^2 - 5x & x > 1. \end{cases}$$

Both formulas give $y = -3$ at $x = 1$, so the curve is continuous there and passes through $(1, -3)$.

The point $(1, -3)$**.** It is not a stationary point: the gradient from the left is $6x^2 - 5 = 1$ at $x = 1$, and from the right it is $4x - 5 = -1$. More precisely,

- for $\sqrt{5/6} < x < 1$ we have $\mathrm{d}y/\mathrm{d}x = 6x^2 - 5 > 0$, so y increases up to -3;
- for $1 < x < \frac{5}{4}$ we have $\mathrm{d}y/\mathrm{d}x = 4x - 5 < 0$, so y decreases from -3.

So $y < -3$ at all points near $x = 1$ on either side, and $(1, -3)$ is a local maximum, a corner of the curve.

The other turning points. On $x < 1$, $\mathrm{d}y/\mathrm{d}x = 6x^2 - 5 = 0$ gives $x = \pm\sqrt{5/6} = \pm\frac{\sqrt{30}}{6}$, both less than 1. Using $2x^2 = \frac{5}{3}$,

$$y = x(2x^2 - 5) = -\tfrac{10}{3}x = \mp\tfrac{10}{3} \cdot \tfrac{\sqrt{30}}{6} = \mp\tfrac{5\sqrt{30}}{9}.$$

Since $\mathrm{d}^2y/\mathrm{d}x^2 = 12x$, the point with $x < 0$ is a maximum and the one with $x > 0$ a minimum. On $x > 1$, $4x - 5 = 0$ gives $x = \frac{5}{4}$, with $y = \frac{25}{8} - \frac{25}{4} = -\frac{25}{8}$, a minimum since the parabola opens upwards. So the other three are

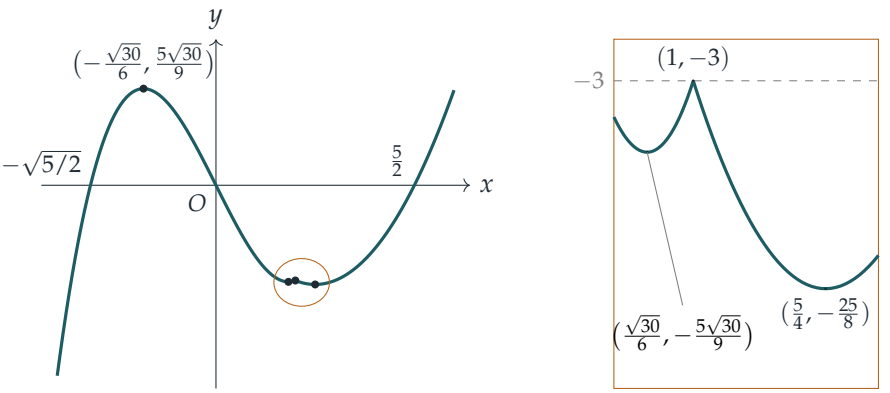
$$\text{maximum } \left(-\tfrac{\sqrt{30}}{6},\ \tfrac{5\sqrt{30}}{9}\right), \qquad \text{minima } \left(\tfrac{\sqrt{30}}{6},\ -\tfrac{5\sqrt{30}}{9}\right) \text{ and } \left(\tfrac{5}{4},\ -\tfrac{25}{8}\right).$$

Sketch. The cubic $2x^3 - 5x$ is odd, so the part of the curve with $-1 \le x \le 1$ has rotational symmetry about the origin. It crosses the axes where $x(2x^2 - 5) = 0$ with $x \le 1$, giving $x = 0$ and $x = -\sqrt{5/2} \approx -1.58$, and where $x(2x - 5) = 0$ with $x > 1$, giving $x = \frac{5}{2}$. As $x \to -\infty$, $y \to -\infty$; as $x \to \infty$, $y \to \infty$.

The three points near $x = 1$ are very close together, so it is worth deciding their order. Squaring the positive numbers $\frac{5\sqrt{30}}{9}$ and $\frac{25}{8}$ gives $\frac{750}{81} \approx 9.26$ and $\frac{625}{64} \approx 9.77$, so

$$-\tfrac{25}{8} < -\tfrac{5\sqrt{30}}{9} < -3 :$$

the minimum at $x = \frac{5}{4}$ is the lower one, and both minima are below the corner. On a sensible scale this detail is invisible, so the inset magnifies it.



Checks

- (ii): at $x = \frac{3}{2}$, $\mathrm{Min}(\frac{9}{4}, 3) = \frac{9}{4}$ and $2 \cdot \frac{9}{4} = \frac{9}{2} = 5 \cdot \frac{3}{2} - 3$; at $x = 3$, $\mathrm{Min}(9, 6) = 6$ and $12 = 15 - 3$.
- (iii), $m = 6$: try $x = -1$, where $\mathrm{Min}(1, -2) + \mathrm{Min}(-1, -4) = -2 - 4 = -6 = 6x$; and $x = 1$, not in the set, gives $1 + 1 = 2 \neq 6$.
- (iv): a computer scan of y at steps of 10^{-5} on $[-3, 3]$ found exactly four turning points, at $x \approx -0.9129$ ($y \approx 3.0429$), 0.9129 (-3.0429), 1 (-3) and 1.25 (-3.125), matching the values above.

Common slips

- Keeping roots that fall outside the interval in which their formula applies, such as $x = -2$ in (iii) with $m = 2$.
- Solving $6x = 6x$ as if it gave a single value, instead of recognising that it holds on the whole interval.
- Smoothing the curve in (iv) at $(1, -3)$: it has a genuine corner there, with gradient 1 on the left and -1 on the right.

Beyond STEP

Functions like $\mathrm{Min}(f,g) = \frac{1}{2}\big(f+g-|f-g|\big)$ are always continuous when f and g are, but they need not be differentiable, and (iv) shows that a local maximum need not be a stationary point. In analysis courses this leads to the careful statement of the result that extrema of a *differentiable* function occur where the derivative vanishes, and to one-sided derivatives. Piecewise-defined functions of this kind are also central in optimisation, where the minimum of several linear functions gives the corners studied in linear programming. Abbott's *Understanding Analysis* treats continuity and differentiability with exactly this sort of example in mind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.